Research Status, Contribution and Completion Plan

Hierarchical Federated Learning for Privacy-Aware, Low-Latency Multimodal IoT Systems

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1. What this research is

A hospital cannot pool patient records, and federated learning solves only half of that problem: it moves the model instead of the data, but the model updates themselves leak. Differential privacy closes the leak, and here the work meets a question the field has largely stepped around. **A privacy guarantee is stated over a unit, and everybody states it over the training record, while the unit a hospital actually owes its duty to is the person.** Where one episode of care produces many records — and in an ICU it produces dozens — those two units are not the same, for every single patient.

This research builds a three-tier hierarchical federated system in which the privacy unit is the patient *by construction*, and then measures what that choice costs. The system is validated end to end — network, compute, energy and deployed inference — so the result is a deployable system rather than an accuracy number.

The question the research answers: *what does a patient-level privacy guarantee cost, what actually buys it, and does that cost transfer between hospitals? Nobody in the cited literature has answered it.*

2. What has been proved

Every figure below is read from an artefact under `SIMULATORS/results/`. Two corpora are reported: VitalDB (surgical ICU, 9 constructed clients) and WEAR (wearable, 22 clients of one participant each).

2.1 The system meets its design objectives

#	Objective	Target	VitalDB	WEAR	Verdict
O1	Uplink traffic reduction	≥ 50%	90.8%	96.1%	PASS
O2a	Privacy budget (composed)	$\varepsilon \le 1$	0.970	0.966	PASS
O2b	Accuracy loss under DP	≤ 2 pp	−0.92 pp	1.967 pp	PASS / INCONCLUSIVE
O3a	Inference latency (P95)	< 100 ms	3.05 ms	1.22 ms	PASS
O3b	Successful model updates	> 99%	100%	99.94%	PASS
O4	Energy saving per round	≥ 20%	72.3%	76.8%	PASS

The composed budget is the honest one: it includes the cost of releasing the conformal calibration quantile, which comparable work omits. WEAR's O2b spans the 2 pp threshold ([1.760, 2.173]) and is reported as undetermined rather than extended until it passed.

2.2 Findings

	Finding	Evidence	Status
F1	**Patient-level DP costs 1.77× the noise, not 8× the budget.** Converting a record-level guarantee to a patient-level one after the fact degrades $\varepsilon \approx 1$ to $\varepsilon \approx 8$ at unusable δ. Direct accounting is dramatically tighter.	σ 0.9985 → 1.7676	**PROVED**
F2	**Pretrained initialisation, not head-only DP, is what makes a strict budget survivable.** Isolated by a B4/B7/B6 decomposition. Head-only DP adds ~0 accuracy; its value is communication. This refutes the stated rationale of our own mechanism.	19× / 79× less privacy-sensitive; monotone 5/5 both corpora	**PROVED**
F3	**The cost of the privacy unit grows with records per individual** and vanishes at k=1, where a patient *is* a record — so the experiment carries its own positive control.	gap 0.0507 → 0.0832 → 0.0866 over k=2,4,8; k=1 null (p=0.75)	**PROVED** (VitalDB)

	Finding	Evidence	Status
F4	**Compression is free.** Sparsification and 8-bit quantization are well-powered nulls, so the uplink saving costs no accuracy.	paired ablation, null	**PROVED**
F5	**The predicted patient-level leakage advantage does not exist.** A LiRA membership-inference audit at the individual level, with a passing positive control, refutes the effect the design was expected to show.	bounds a large effect only (±0.060)	**REFUTED**
F6	**Adaptive per-layer privacy allocation is worse than a uniform budget.** Composition across groups charges more, so every group must be noisier.	measured, reported as a negative	**NEGATIVE**
F7	**The hierarchy costs accuracy and buys communication.** About –0.005 AUROC, resolved and replicated. It must never be presented as an accuracy benefit.	–0.0047 (p=.032) / –0.0052 (p=.0018)	**RESOLVED**

3. What the external-validation phase discovered

The largest objection to this work was that every privacy claim rested on one dataset from one hospital, whose clients were a partition we invented. eICU-CRD answers it: 3,782,805 windows, 132,736 individuals, **187 real hospitals taken as given**, a median of 444 individuals per client. Its encoders were pretrained on a population verified disjoint from every client, so — unlike every VitalDB figure — its utilities carry no oracle-configuration caveat.

	Result	Reading
D1	**The positive control passed.** At k=1 the arms collapse as the construction requires (gap –0.0107, inside the ±0.020 tolerance fixed in advance).	Harness validated; all other eICU results admissible.
D2	**The k-law's magnitude did not transfer.** Observed gaps are ~1/6 of predicted; 3 of 4 fall outside the ±0.030 band. The *shape* is preserved (monotone rise –0.0107, +0.0060, +0.0110, +0.0135 over k=1,2,4,8).	P1b refuted. Registered reading: shape structural, magnitude corpus-dependent.
D3	**Nothing resolves at three seeds.** All CIs cover zero, all p > 0.12; ~49 seeds would be needed. Seeds were fixed in advance and are not extended after seeing the data.	Directionally consistent, not replicated.
D4	**Real-hospital heterogeneity is narrower than constructed** (0.0–18.3% vs 3.2–50.7%), decided from the corpus with no training.	P4a refuted; removes P4b's rationale.
D5	**Fusion is null on eICU** (+0.0037 AUROC, 0.2× seed-sd, AUPRC –0.0001) — on the corpus the Discussion itself nominated as the fair test.	The multimodal gain claim does not generalise.
D6	**A training-collapse defect was found and fixed.** On long epochs the model peaks near step 1,000 then decays to constant output; validation ran once per 41,394-step epoch. It also affected encoder pretraining, whose output k-scaling *freezes*.	Caught before it silently poisoned every eICU number.
D7	**The headroom explanation for D5 was tested and refuted.** eICU has MORE headroom than VitalDB (0.1788 vs 0.1717) and captures a tenth as much of it (2.1% vs 21.5%).	Withdrawn. Ceiling effects explain WEAR, not eICU.

Two corrections belong in the record: eICU yields 206 usable hospitals and 187 clients, not the 208 quoted in the planning documents, and observed mortality is 0.0422 against a pre-registered forecast of 0.0260.

4. N-D — the one item that adds an original mechanism

The dossier calls N-D *"the only remaining candidate for an original mechanism rather than a measurement of an existing one."* Implemented 20 September 2026; the experiment is queued. It matters more than the dossier itself recorded.

The observation. In `model/privacy/patient_dp.py` a patient's gradient is summed over that patient's k windows and then clipped to a fixed norm C. The pre-clip norm therefore grows with k, so patients with long recordings saturate the clip and are systematically down-weighted. The module's own docstring already noted this as a cost.

Why it bears on the headline. The k-scaling law reports that the patient arm is FLAT in k while the record arm rises, and concludes that extra records buy the patient arm nothing. If saturation causes that flatness, the paper's central cost figure is partly an artefact of a clipping choice rather than an inherent property of the guarantee.

The intervention. Rescale the summed gradient by k or √k *before* the clip. Arithmetic check run before the experiment, C=1, pre-clip norm proportional to k: under `none` the clip binds from k=8 upward; under `k` it never binds; under `sqrt_k` only at k=32.

The guarantee is unchanged, and this must be stated precisely or the mechanism reads as a privacy cheat. Sensitivity is set by the clip, not by what is clipped: after rescaling, every patient's contribution is still bounded by C, so removing one patient still moves the sum by at most C. Sigma, epsilon and delta are identical across arms — verified by inspection, since `clip_norm` appears in no noise-multiplier or accountant code path. The arms compare at a matched budget and the difference is optimisation geometry alone. This is the same argument the paper already makes for head-only DP, and it has to be made the same way here.

Frozen predictions (`PREREGISTRATION_ND_CLIPNORM.md`): N1 clip saturation >0.50 under *none* and <0.10 under *k*; N2 patient-arm AUPRC rises by ≥0.02; N3 the k=8 gap shrinks by ≥0.02. If N1 holds and N2 fails, the cost is inherent to the guarantee, which *strengthens* the existing claim — that outcome is as reportable as success.

5. Contribution, and how unique it is

Measured against the five works this paper positions itself against, the gap is specific. **None asks what the privacy unit should be when one person contributes many records, and none measures what choosing correctly costs.**

Prior work	What it does	What it does not do
FL2DP (Gu et al., IEEE TII 2024)	DP with shuffling, exponential mechanism	Record-level unit; no hierarchy
HED-FL (De Rango et al., 2023)	Hierarchical, energy-efficient FL	No privacy guarantee at all
HCEF (Zhang et al., IEEE TMC 2025)	Heterogeneity-aware, compression	No patient-level accounting
RoPPFL (Zhou et al., 2024)	Local DP, poisoning defence	Record-level; no edge tier in the budget
Mhaisen et al. (IEEE TNSE 2022)	Statistical user–edge assignment	No privacy unit analysis

C1 — The patient as the privacy unit, with the edge tier inside the accounting. Gradients grouped by individual before bounding, sampling over individuals, noise calibrated against every client so the edge tier is reflected in the reported budget. Cost: 1.77× noise, a number that does not exist in the literature for multimodal clinical FL.

C2 — The cost of the unit as a measured law with a built-in positive control. At k=1 a patient is a record, so the arms must coincide; they do, on both corpora. A self-validating design is a methodological contribution and it is what makes the second corpus's miss interpretable rather than ambiguous.

C3 — A boundary condition established against a dated forecast. Shape transfers, magnitude does not. Because the prediction was frozen before the corpus existed, this reads as a result rather than a rationalisation.

C4 — A mechanism, not only a measurement (N-D, pending). If the cost is partly an artefact of fixed-norm clipping, the paper moves from *what a patient-level guarantee costs* to *what it costs, why, and how to reduce it.*

C5 — A per-task criterion for multimodal fusion (pending). Fusion is worth +0.0370 AUROC on VitalDB, untestable on WEAR, null on eICU. The obvious explanation — headroom — was tested and refuted, so the criterion must be named and tested rather than asserted.

6. Q1/Q2 quality — a candid assessment

Strengths a reviewer will credit	Weaknesses a reviewer will attack
Three pre-registrations, two predictions already falsified (P1b, P4a). Almost no submission at this tier has this.	**Statistical power.** The eICU k-law resolves nothing at n=3; the ε sweep is single-seed per point; a component was dropped on a one-seed comparison.

Strengths a reviewer will credit	Weaknesses a reviewer will attack
Positive controls throughout — the k=1 collapse, the LiRA memorising configuration. Nulls are interpretable rather than ambiguous.	**Several arms do not resolve.** Only C2 resolves in the component ablation and its sign flips between corpora, so component necessity is not established.
Three corpora, one of them 187 real hospitals with end-to-end-valid disjoint pretraining.	**Oracle encoders on VitalDB.** Every VitalDB utility is an upper bound; the deployable configuration costs 0.0322 AUROC. eICU fixes this; VitalDB does not.
End-to-end validation: packet-level network, energy, ONNX deployment — not an accuracy table.	**Fusion is null on the fair-test corpus**, so the multimodal premise is supported conditionally at best.
Negatives reported rather than dropped, including against the authors' own mechanism, and a withdrawn explanation (headroom) recorded in writing.	**One objective is undetermined** (WEAR O2b) and must stay that way.

Honest verdict. The contribution is real and the gap in the literature is genuine, so the work is a credible fit for IEEE TII or IoT-J. The decisive factor will not be novelty — it is whether the honesty reads as rigour or as weakness. Framed as *"we measured what a patient-level guarantee costs, proved the harness with a control, and found the magnitude is cohort-specific"* it is a strong paper. Framed as *"our system wins"* the unresolved arms and the fusion null will sink it. If N-D succeeds the paper gains an original mechanism and moves up a class.

7. Complete status register

7.1 Done

Area	Item
Infrastructure	Phases 1–6; NS-3 packet-level; CloudSim Plus; ONNX deployment
Objectives	O1–O4 — VitalDB 7 PASS; WEAR 6 PASS (O2b undetermined, by design)
Findings	Patient-level DP 1.77× noise · pretrained init 19×/79× · k-law on VitalDB with k=1 control · compression free · MIA refuted · SADP negative · hierarchy resolved (–0.005 AUROC, buys uplink)
eICU	Corpus built (3.78M windows, 187 hospitals) · Gate 1 validate PASS · Gate 2 kill switch PASS · P1a control PASS · P1b refuted (k=1,2,4,8) · P4a refuted · fusion null at 5 seeds · RIGOROUS disjoint encoders
Fusion test	Headroom hypothesis refuted · balance treatment complete (gain –0.0062; F1 held, F2/F3 failed — lever overshot)
Engineering	Training-collapse fix in 3 trainers (opt-in, published results unchanged) · eICU label-conflict → `metric_group_file` · k_scaling validity message · 9 argparse gates · `--feat_b_zero` · **N-D** `--clip_norm` **implemented**
Dossier items	Fig. 4 redrawn from measured LiRA scores · ablations A1–A4 (both were stale in the dossier)
Governance	Three pre-registrations · amendments A1 and F-A1 recorded
Writing	`results_kscaling_eicu.tex` drafted · `abstract_draft.tex` + keywords drafted · this document

7.2 Running — about 11–12 hours, unattended

Job	Progress	Remaining	What it settles
k-scaling k=16 (eICU)	26/30 arm-runs	~8–9 h	P1c saturation — the one prediction VitalDB cannot make
Fusion matched control	12/15 arms	~20 min	Makes the balance test interpretable (guards a 20% partition change)
N-D clip-norm (VitalDB)	queued	~2–3 h	Whether the headline cost is partly a clipping artefact

7.3 Not done

Priority	Item	Effort	Why it matters
1	**Four Discussion subsections** — privacy-unit cost, component necessity, position vs prior art, deployment	2–3 days	Unwritten; the largest single gap to submission

Priority	Item	Effort	Why it matters
2	**Conformal coverage validation**	~3 h	ε_{conf}=0.20 is a fifth of the reported budget and its coverage has never been measured
3	**Multi-seed B4/B6/B7**	~1 day	The 19×/79× headline is single-seed, with no intervals
4	Merge abstract; reconcile fusion subsection with the eICU null	1 day	Drafted
5	Correct 208→187 and the stale 2-seed fusion figures in `docs/Research.md`	0.5 day	Mechanical, but they will reach a draft otherwise
6	§28.1 rigorous pretraining for VitalDB/WEAR	~1 day	Optional — eICU already carries the valid claim
—	S3 hierarchical-no-DP; mask-channel ablation	—	Optional; drop unless a reviewer asks

8. Where this ends

The experiments are finished or queued; nothing further needs to be designed. The binding constraint from here is writing, not compute.

Estimate: about two weeks to a submittable manuscript, of which roughly half a day is compute the machine runs by itself and the remainder is writing and verification.

Definition of done. Seven sections written; every number traceable to a file under `SIMULATORS/results/`; no placeholder text anywhere; all three pre-registrations and both amendments disclosed; N-D, conformal coverage and the fusion verdict reported whatever their outcome; and no claim in the paper that the authors' own data contradicts.

What would be scope creep: *a fourth corpus, further ablations beyond the three listed, or raising seed counts on an already-registered test after seeing its result — the last would convert a pre-registered test into a search and destroy the paper's principal methodological asset. The research ends when k=16, the fusion control and N-D return their verdicts, whatever those verdicts are.*